MODULE 3 MUSCLE STRUCTURE

Muscle Structure & Sarcomeres.

MUSCLE STRUCTURE AND SARCOMERES

A reference for the muscle structure video and lab. This chapter covers the structure of a skeletal muscle from its connective tissue coverings, down through the muscle fiber, to the sarcomere and its myofilaments. The contraction mechanism belongs to physiology and is not covered here.

BY THE END

1. Name the connective tissue coverings of a skeletal muscle and trace how they connect to the tendon.
2. Order the levels of muscle organization, from the whole muscle down to the myofilament.
3. Identify the parts of a muscle fiber, including the T tubules, sarcoplasmic reticulum, and triad.
4. Diagram a sarcomere, name its bands, lines, and zones, and state which filaments lie in each.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Muscle structure

drsrennie-stack.github.io/new-build-bio4-solano/muscle-structure-concept-videos.html

Skeletal Muscle, an Overview

A skeletal muscle is an organ. It is built of muscle tissue plus connective tissue, blood vessels, and nerves, and it is wrapped, bundled, and anchored at every level of its structure.

THE MUSCLE AS AN ORGAN

Structure	What it is
Skeletal muscle	A voluntary, striated muscle organ, usually attached to bone, that produces movement when it contracts
Muscle fiber	A single skeletal muscle cell, long and cylindrical, with many nuclei
Origin	The attachment of a muscle to the more stationary bone
Insertion	The attachment of a muscle to the more movable bone, the bone that is pulled toward the origin
Belly	The thick, fleshy middle region of the muscle between its attachments

Attachments and Action

A muscle crosses at least one joint and attaches to the articulating bones. When it contracts, one bone stays put and the other moves.

ORIGIN, INSERTION, ACTION

Structure	What it is
Origin	The tendon's attachment to the stationary bone, usually the proximal one
Insertion	The tendon's attachment to the movable bone, usually the distal one, pulled toward the origin
Action	The movement the muscle produces when it contracts

How Muscles Work Together

Skeletal muscles are arranged in opposing pairs across a joint, and each contraction has muscles playing different roles.

THE ROLES IN A MOVEMENT

Structure	What it is
Agonist (prime mover)	The muscle that contracts to produce the action
Antagonist	The opposing muscle that stretches and yields to the prime mover
Synergist	A muscle that aids the action or prevents unwanted movement
Fixator	A muscle that stabilizes the origin so the prime mover acts efficiently
Compartment	A group of muscles, vessels, and nerves with a common function, such as the flexor and extensor compartments

How Skeletal Muscles Are Named

A muscle's name usually encodes one or more of these features. Learn the clues and the name tells you the muscle.

NAMING CLUES AND WHAT THEY LOOK LIKE IN A NAME

Naming clue	Examples in the name
Direction of fibers	rectus, transverse, oblique
Size	maximus, minimus, longus, brevis, vastus
Shape	deltoid, trapezius, serratus, rhomboid, orbicularis
Action	flexor, extensor, abductor, adductor, levator, pronator, supinator
Number of origins	biceps, triceps, quadriceps
Location	temporalis, over the temporal bone
Origin and insertion	sternocleidomastoid, sternum and clavicle to the mastoid process

The Three Muscle Tissue Types

All three are muscle, but their structure differs. Compare them by location, appearance, control, and cell features.

LOCATION, CONTROL, AND THE CELL

Type	Location	Striations and control	Cell features
Skeletal	Attached to the skeleton	Striated, voluntary	Long cylindrical fibers, many nuclei pushed to the edge
Cardiac	The heart wall	Striated, involuntary	Branching fibers, one central nucleus, joined by intercalated discs with desmosomes for adhesion and gap junctions for electrical coupling
Smooth	Walls of hollow organs and vessels	Non-striated, involuntary	Spindle-shaped cells, one central nucleus

Connective Tissue Coverings

Three connective tissue sheaths wrap the muscle at three scales. They are continuous with each other and with the tendon, so a fiber's pull is carried all the way to the bone.

THE THREE SHEATHS

Sheath	What it wraps	Tissue
Epimysium	The entire muscle, the outermost layer	Dense irregular connective tissue
Perimysium	Each fascicle, a bundle of muscle fibers	Dense irregular connective tissue
Endomysium	Each individual muscle fiber	A fine sheath of areolar connective tissue with reticular fibers

FASCIA AND THE ATTACHMENTS THEY FORM

Structure	What it is
Superficial fascia	The subcutaneous layer of areolar and adipose tissue between the skin and the muscles
Deep fascia	A sheet of dense connective tissue that wraps muscles, separates them, and binds those with a common action into functional groups
Tendon	A cord of dense regular connective tissue, the merged continuation of all three sheaths, that attaches a muscle to bone
Aponeurosis	A broad, flat sheet of tendon-like connective tissue, an attachment used where a muscle is wide

Levels of Organization

A skeletal muscle is a set of structures nested inside one another. Follow them in order, from the whole organ down to the protein strands.

1. **Muscle** the whole organ, wrapped in epimysium
2. **Fascicle** a visible bundle of muscle fibers, wrapped in perimysium
3. **Muscle fiber** a single muscle cell, wrapped in endomysium
4. **Myofibril** a long contractile rod running the length of the fiber; hundreds to thousands fill each fiber
5. **Myofilament** the thick and thin protein strands within a myofibril; their ordered overlap forms the sarcomere

The Muscle Fiber

The muscle fiber is the muscle cell. Its parts carry the names of ordinary cell structures with the prefix sarco, from the Greek for flesh.

INSIDE THE FIBER

Structure	What it is
Sarcolemma	The plasma membrane of the muscle fiber
Sarcoplasm	The cytoplasm of the muscle fiber
Myonuclei	The many nuclei of the fiber, flattened against the inner sarcolemma; a muscle fiber is multinucleated
Myofibrils	The contractile rods that fill most of the sarcoplasm
Transverse (T) tubules	Narrow tunnels of the sarcolemma that dive inward, carrying the surface signal deep into the fiber
Sarcoplasmic reticulum	The smooth endoplasmic reticulum of the muscle fiber, a network of tubules wrapping each myofibril; it stores calcium ions
Terminal cisternae	The enlarged end sacs of the sarcoplasmic reticulum, set on either side of a T tubule
Triad	The structural unit of one T tubule flanked by two terminal cisternae
Mitochondria	Packed in rows between the myofibrils, they supply the ATP that contraction demands
Myoglobin	A red, oxygen-binding pigment in the sarcoplasm that stores oxygen and gives muscle its color
Glycogen	Granules of stored carbohydrate fuel held in the sarcoplasm

Myofilaments, Thick and Thin

Two kinds of protein filament fill each myofibril. Compare them by their main protein and structure.

THE TWO MYOFILAMENTS

Myofilament	Main protein	Structure	Anchored at
Thick filament	Myosin	A bundle of myosin molecules; each molecule has a long tail and a pair of heads, and the heads project outward as cross bridges	The M line
Thin filament	Actin	Two twisted strands of actin subunits; each subunit carries a myosin-binding site	The Z disc

THE TWO REGULATORY PROTEINS

Structure	What it is
Tropomyosin	A rod-shaped regulatory protein that lies end to end along the thin filament
Troponin	A three-part regulatory protein bound at intervals to tropomyosin on the thin filament

The Sarcomere

The sarcomere is the contractile unit of muscle, the segment of a myofibril from one Z disc to the next. Its bands and zones are named for how the overlapping filaments look under the microscope.

COMPARE THE REGIONS

Region	What it is	Filaments present
Z disc	The protein plate at each end of the sarcomere, its border	Anchors the thin filaments
A band	The dark band, as wide as the thick filaments are long	Thick filaments across its full width, with thin filaments overlapping at each edge
I band	The light band; a Z disc runs through its center, so one I band is shared by two sarcomeres	Thin filaments only
H zone	The lighter region in the center of the A band	Thick filaments only
M line	The protein line in the center of the H zone	Anchors the thick filaments to one another
Zone of overlap	The region at each end of the A band	Both thick and thin filaments, interleaved

WHAT YOU SEE ON THE SLIDE

Structure	What it is
Striations	The stripes of skeletal muscle, the alternating dark A bands and light I bands of thousands of aligned sarcomeres

Structural Proteins of the Sarcomere

Besides the contractile proteins, a set of structural proteins holds the sarcomere in register and links it to the cell membrane. Compare them by location and role.

LOCATION AND ROLE

Protein	Location	Role
Titin	Spans from the Z disc to the M line	Anchors the thick filament, gives the sarcomere elastic recoil, and keeps it from overstretching
Nebulin	Runs the length of the thin filament	Anchors the thin filament and sets its length
Alpha-actinin	The Z disc	Binds the thin filaments and titin to the Z disc
Myomesin	The M line	Binds the thick filaments together at the M line
Dystrophin	Links the thin filaments to the sarcolemma	Transfers the force of contraction from the sarcomere out to the connective tissue; faulty in muscular dystrophy

KEY POINT

The A band never changes length, because it is the thick filament. The I band and the H zone are the ones that narrow. Any question about band width comes back to that.

STRUCTURE EXPLAINS THE INJURY

The three connective tissue sheaths are continuous with one another and with the **tendon**, so the pull of every fiber reaches the bone. This continuity also explains injury. A muscle strain tears fibers and endomysium within the belly, while a tendon rupture separates the muscle from the bone entirely, and the two heal on very different timelines.

Muscular dystrophy is a group of inherited diseases of muscle structure. In Duchenne muscular dystrophy the gene for **dystrophin** is faulty, so the link between the sarcomere and the sarcolemma fails, the fiber damages itself with ordinary use, and weakness is progressive.

The **triad**, one T tubule flanked by two terminal cisternae, is the structural meeting point where a surface signal reaches the calcium store of the sarcoplasmic reticulum. Its tight, repeating arrangement is what lets an entire fiber respond almost at once.